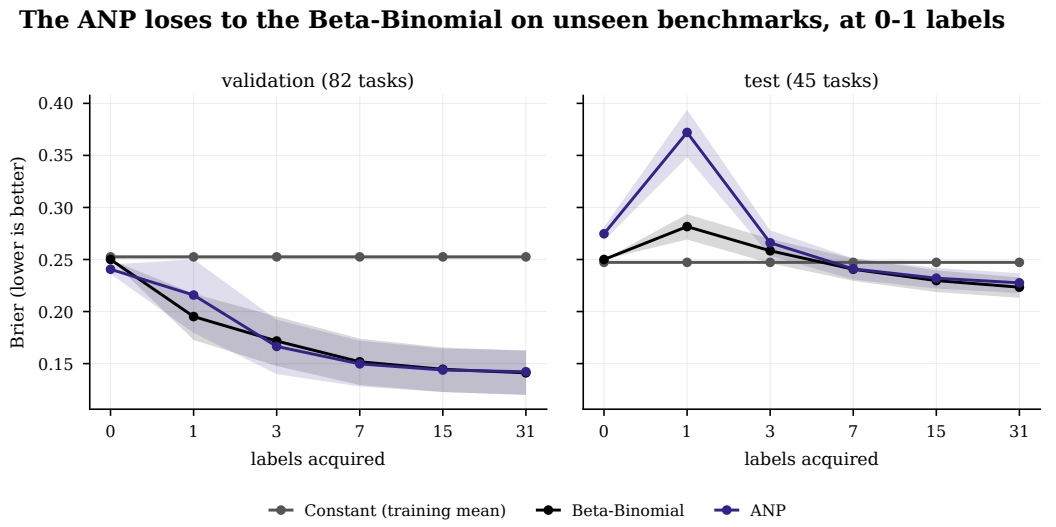


Figure (single-pair). Mean Brier on each task's held-out evaluation items after 0-31 labels, each subject-benchmark task simulated alone on its own labels with random acquisition (validation: 3 candidate orders, test: 10; benchmarks never seen in training). Bands: 95% bootstrap over tasks. The ANP's whole deficit sits at 0 and 1 labels; from 3 labels on it matches the Beta-Binomial. The constant ignores labels and sets the floor. Brier ALC (val / test): Constant (training mean) 0.2525 / 0.2472; Beta-Binomial 0.1717 / 0.2494; ANP 0.1735 / 0.2725. Pooled Rasch needs other pairs' labels and is shown on the multi-pair page.

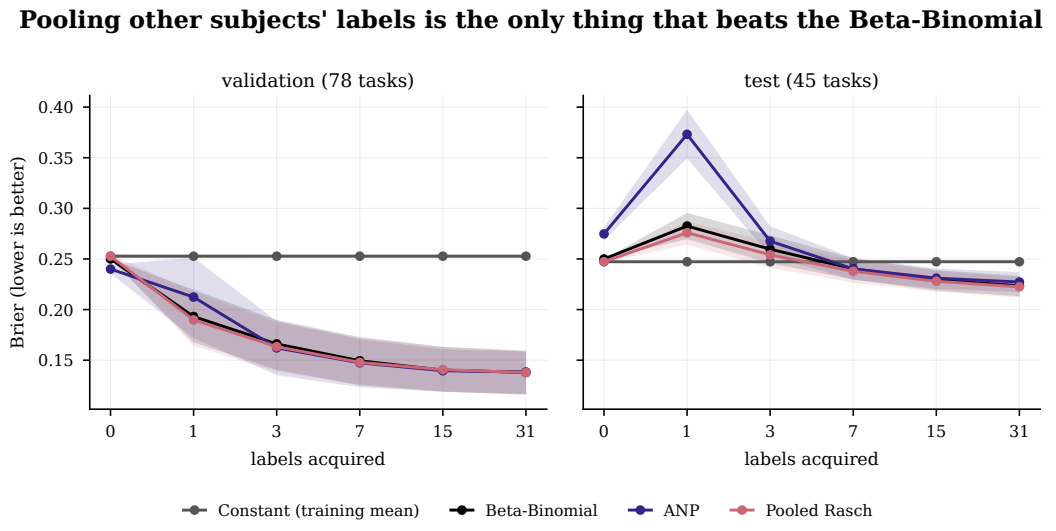


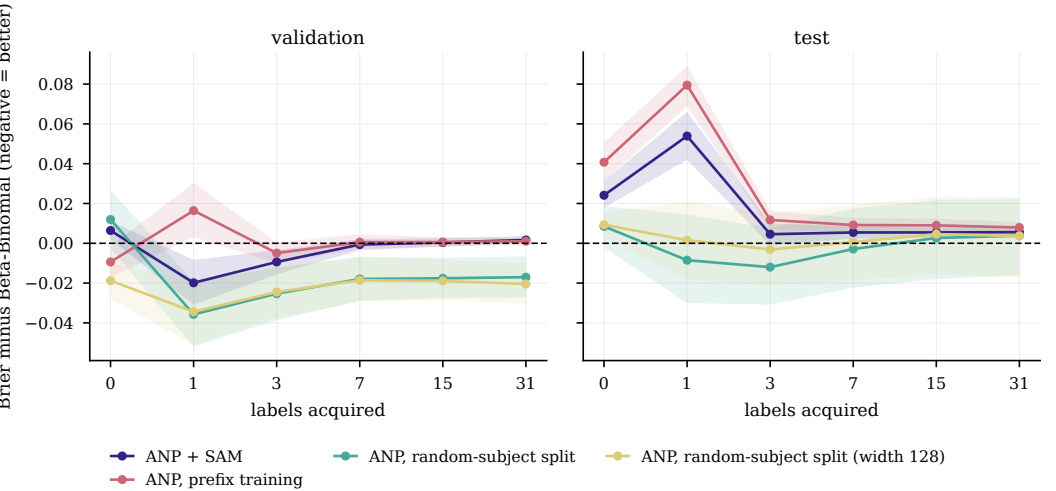
Figure (multi-pair). Same models on CodaBench-like draws: one or two subjects per benchmark, labels acquired in the the online simulation's frozen orders, and every pair's labels available to the predictor (60 validation and 150 test draws). Models that ignore other pairs are placed on the same draws from their single-pair scores. Pooled Rasch fits benchmark rate, subject ability and item difficulty from all pairs' labels and is the only model below the Beta-Binomial on test at every budget. Bands: 95% bootstrap over tasks. Brier ALC (val / test): Constant (training mean) 0.2527 / 0.2472; Beta-Binomial 0.1685 / 0.2497; ANP 0.1701 / 0.2726; Pooled Rasch 0.1674 / 0.2461.

Only pooling other subjects' labels beats the Beta-Binomial on unseen bench

model	protocol	val ALC	test ALC	test minus Beta-Binomial [95% CI]
ANP training recipe and split				
ANP + SAM	single-pair	0.1667	0.2663	+0.0168 [+0.0128, +0.0211]*
ANP, prefix training	single-pair	0.1735	0.2762	+0.0267 [+0.0231, +0.0306]*
ANP, random-subject split	single-pair	0.1519	0.2465	-0.0029 [-0.0209, +0.0154]
ANP, random-subject split (width 128)	single-pair	0.1485	0.2515	+0.0020 [-0.0153, +0.0190]
Combining the ANP with the Beta-Binomia				
ANP	single-pair	0.1735	0.2725	+0.0230 [+0.0198, +0.0267]*
ANP with Beta-Binomial fallback at 0-1 labels	single-pair	0.1703	0.2519	+0.0025 [+0.0003, +0.0048]*
ANP-Beta-Binomial ensemble	single-pair	0.1702	0.2556	+0.0062 [+0.0050, +0.0074]*
ANP-Beta-Binomial ensemble, per-budget weig	single-pair	0.1688	0.2538	+0.0043 [+0.0027, +0.0062]*
ANP + residual correction	single-pair	0.1936	0.2996	+0.0502 [+0.0465, +0.0537]*
Beta-Binomial + residual correction	single-pair	0.1953	0.2986	+0.0492 [+0.0452, +0.0529]*
Raw hit rate	single-pair	0.1892	0.2879	+0.0385 [+0.0355, +0.0413]*
Hierarchical Beta-Binomial: subject-metad				
Hierarchical Beta-Binomial	single-pair	0.1703	0.2489	-0.0006 [-0.0027, +0.0017]
Hierarchical Beta-Binomial + provider	single-pair	0.1694	0.2452	-0.0043 [-0.0078, +0.0001]
Hierarchical Beta-Binomial + model name, pro	single-pair	0.1710	0.2445	-0.0049 [-0.0090, +0.0001]
Hierarchical Beta-Binomial + provider, release	single-pair	0.1807	0.2626	+0.0131 [+0.0046, +0.0220]*
Hierarchical Beta-Binomial + model name, pro	single-pair	0.1726	0.2455	-0.0039 [-0.0084, +0.0015]
Pooling all pairs' labels (multi-pair)				
Pooled Rasch	multi-pair	0.1674	0.2461	-0.0035 [-0.0045, -0.0026]*
Pooled Rasch without item difficulty	multi-pair	0.1675	0.2477	-0.0020 [-0.0029, -0.0011]*
Pooled Rasch + provider prior	multi-pair	0.1647	0.2500	+0.0003 [-0.0027, +0.0033]
Not simulated online				
ANP, budget-ladder training	-	-	-	trained, never simulated online
ANP + prior anchor	-	-	-	trained, never simulated online
ANP + SAM, budget-ladder training	-	-	-	trained, never simulated online
ANP without item embeddings	-	-	-	training; step-400 checkpoint only

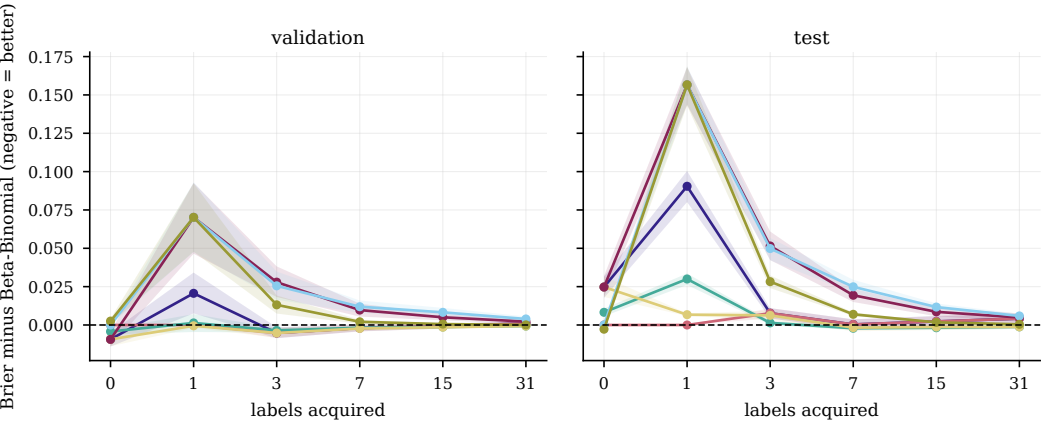
Table. Macro Brier ALC (lower is better) on validation and on the untouched test benchmarks (algotune, cmmlu, editbench), random acquisition. The last column is the paired test difference to the Beta-Binomial (Jeffreys prior) under the same protocol, with a 95% bootstrap interval over tasks; * marks intervals that exclude 0. Single-pair Beta-Binomial: 0.1717 / 0.2494 (val / test); multi-pair: 0.1685 / 0.2497. Validation selected the ensemble weights, the metadata L2 and the pooled-Rasch priors. The random-split ANPs trained on editbench. Rows under 'Not simulated online' were trained but have no online-simulation result.

No training recipe fixes the ANP; having seen the benchmark does



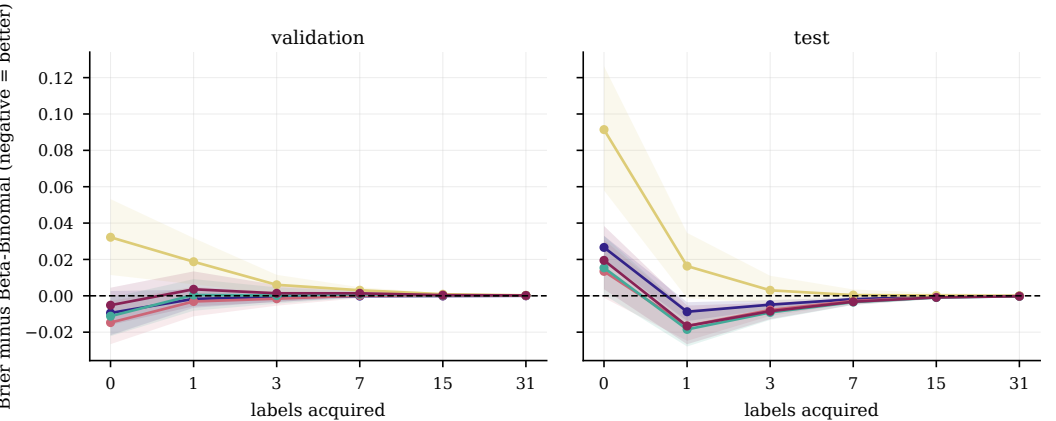
Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. SAM and prefix training of the benchmark-holdout ANP stay above the Beta-Binomial on test, by a wide margin at 0-1 labels and slightly from 3 labels on. The random-split ANPs trained on editbench, one of the three test benchmarks; their small test difference averages a large win on editbench with a loss on algotune and cmmlu.

The Beta-Binomial can limit the ANP's damage, but not turn it into a gain



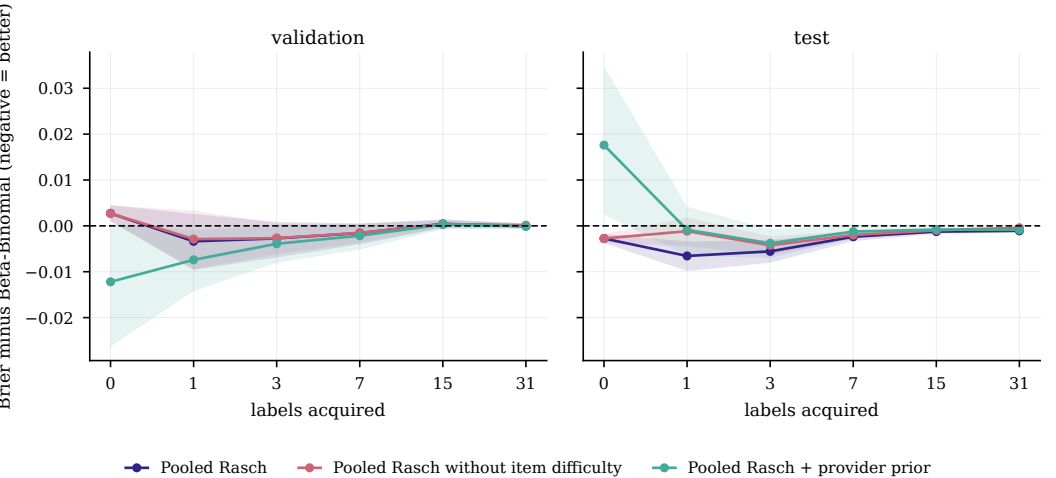
Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. The ANP combined with the Beta-Binomial: the fallback uses the Beta-Binomial below two labels, the ensemble averages the two predictions with weights fitted on validation and frozen for test (the validation curves for the ensembles are in-sample). The residual-correction (AIPW-style) variants shift each prediction by the mean residual of the labels seen so far; after one label that shift is about +/-0.5, which is why they spike at budget 1. The raw hit rate shows the same spike for the same reason: no shrinkage.

Subject metadata barely helps, and release date hurts



Ablation (single-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. A Beta-Binomial with a prior fitted on training benchmarks, whose mean depends on the subject's provider, model name and release date as sent by CodaBench. Test models are newer than any training model, so the release-date term extrapolates and hurts at budget 0. A leave-one-benchmark-out check on the 14 non-test benchmarks did not confirm the small gains of the other arms.

Other subjects' labels help; a provider prior undoes it



Ablation (multi-pair). Paired Brier difference to the Beta-Binomial (Jeffreys prior) at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is the Beta-Binomial. Multi-pair draws (one or two subjects per benchmark). Pooled Rasch fits benchmark rate, subject ability and item difficulty from every pair's labels; removing the item term halves the gain. Centring ability on the subject's provider was chosen on validation but loses on test, at budget 0, because test models are stronger than training models from the same providers.

Test: Brier ALC by model and acquisition policy

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	Random	Uncertainty	Confidence extrem	Bernoulli varia	Expected Brier reducti
Constant (training mean)	**0.2472**	**0.2472**	**0.2472**	**0.2472**	**0.2472**
Raw hit rate	0.2879	0.2881	0.2855	0.2887	0.2860
Beta-Binomial	0.2494	0.2494	0.2478	0.2496	0.2474
ANP	0.2725	0.2710	0.2754	0.2775	0.2727
ANP, prefix training	0.2762	0.2732	0.2835	0.2787	0.2782
ANP + SAM	0.2663	0.2642	0.2718	0.2601	0.2651
ANP, random-subject split*	0.2465	0.2445	0.2471	0.2438	0.2522
ANP, random-subject split (width	0.2515	0.2505	0.2505	0.2544	0.2575

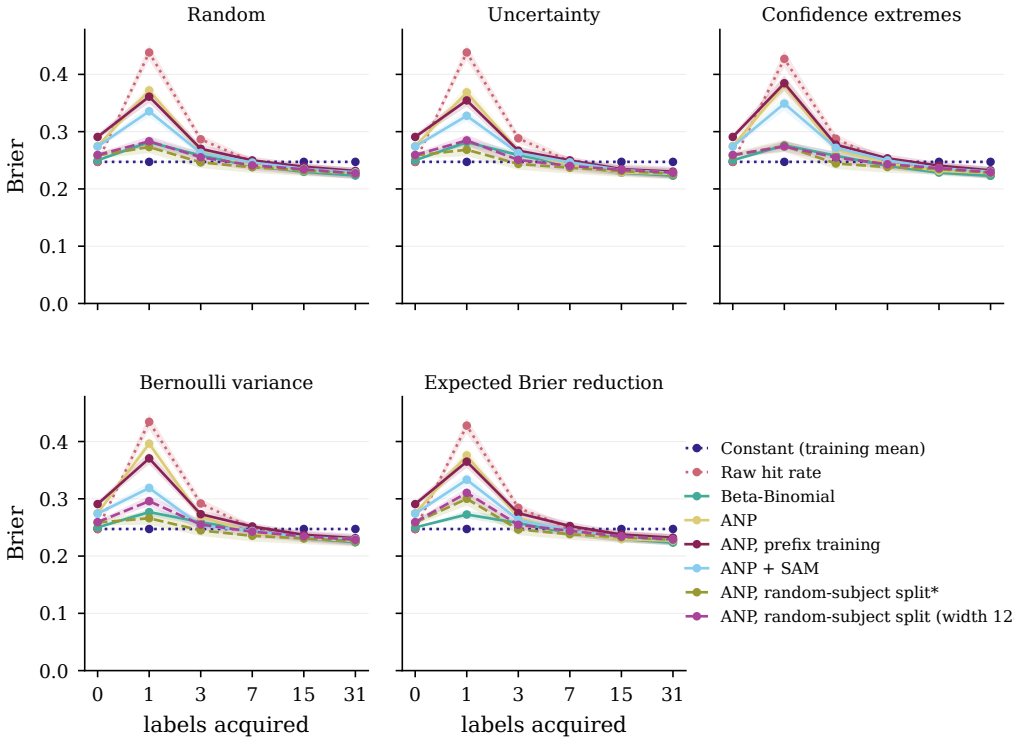
ALC minus Beta-Binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	Random	Uncertainty	Confidence extremes	Bernoulli variance	Expected Brier reductio
Constant (training mean)	-0.0022 ±0.0088	-0.0022 ±0.0091	-0.0006 ±0.0098	-0.0024 ±0.0091	-0.0001 ±0.0099
Raw hit rate	**+0.0385 ±0.0030**	**+0.0387 ±0.0030**	**+0.0376 ±0.0033**	**+0.0390 ±0.0053**	**+0.0386 ±0.0040**
ANP	**+0.0230 ±0.0034**	**+0.0216 ±0.0033**	**+0.0276 ±0.0086**	**+0.0279 ±0.0059**	**+0.0253 ±0.0056**
ANP, prefix training	**+0.0267 ±0.0039**	**+0.0238 ±0.0039**	**+0.0356 ±0.0078**	**+0.0291 ±0.0062**	**+0.0308 ±0.0062**
ANP + SAM	**+0.0168 ±0.0042**	**+0.0148 ±0.0045**	**+0.0240 ±0.0075**	**+0.0105 ±0.0059**	**+0.0177 ±0.0066**
ANP, random-subject split*	-0.0029 ±0.0187	-0.0049 ±0.0197	-0.0007 ±0.0196	-0.0058 ±0.0195	+0.0048 ±0.0192
ANP, random-subject split (width	+0.0020 ±0.0177	+0.0011 ±0.0191	+0.0027 ±0.0188	+0.0047 ±0.0191	+0.0101 ±0.0176

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier by labels acquired (mean ± 1 SE)

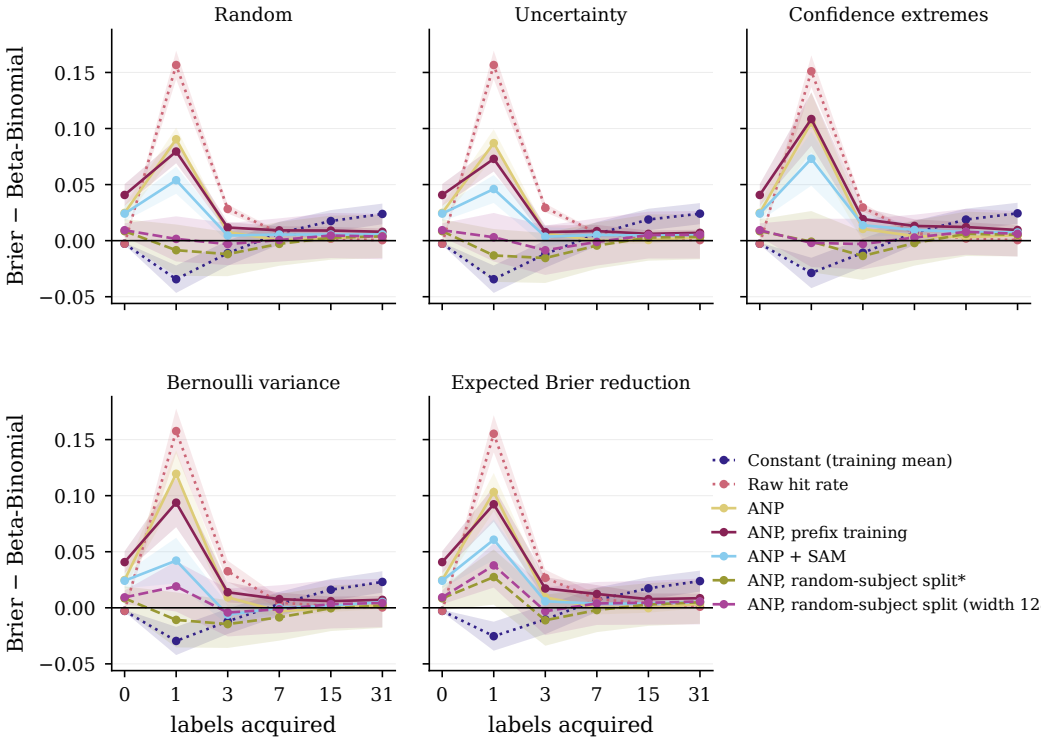
algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier minus Beta-Binomial (paired, 95% CI)

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier ALC by model and acquisition policy

ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	Random	Uncertainty	Confidence extrem	Bernoulli varia	Expected Brier reducti
Constant (training mean)	0.2525	0.2525	0.2525	0.2525	0.2525
Raw hit rate	0.1892	0.1906	0.1944	0.1883	0.1921
Beta-Binomial	0.1717	0.1726	0.1747	0.1704	0.1731
ANP	0.1735	0.1740	0.1759	0.1768	0.1749
ANP, prefix training	0.1735	0.1748	0.1718	0.1729	0.1719
ANP + SAM	**0.1667**	**0.1679**	**0.1677**	**0.1684**	**0.1708**
ANP, random-subject split*	0.1519	0.1538	0.1536	0.1540	0.1611
ANP, random-subject split (width	0.1485	0.1503	0.1521	0.1528	0.1511

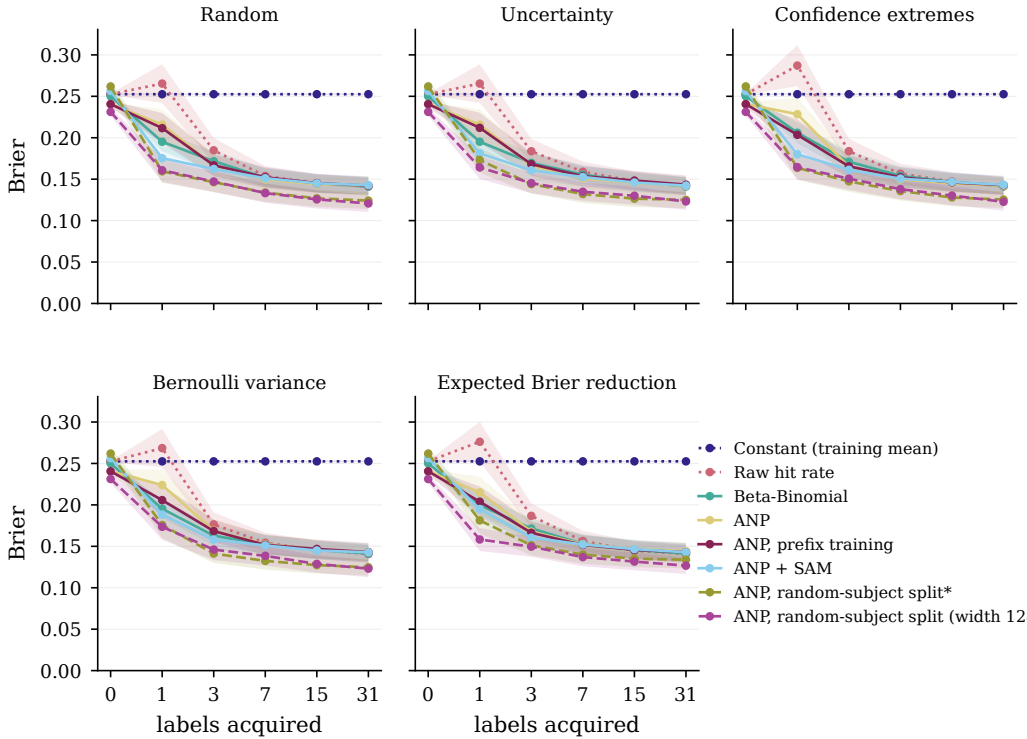
ALC minus Beta-Binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	Random	Uncertainty	Confidence extremes	Bernoulli variance	Expected Brier reductio
Constant (training mean)	**+0.0808 ±0.0211**	**+0.0799 ±0.0213**	**+0.0779 ±0.0214**	**+0.0822 ±0.0210**	**+0.0795 ±0.0212**
Raw hit rate	**+0.0175 ±0.0056**	**+0.0180 ±0.0056**	**+0.0197 ±0.0058**	**+0.0180 ±0.0069**	**+0.0191 ±0.0088**
ANP	+0.0017 ±0.0034	+0.0014 ±0.0034	+0.0012 ±0.0043	**+0.0064 ±0.0060**	+0.0018 ±0.0064
ANP, prefix training	+0.0017 ±0.0044	+0.0022 ±0.0049	-0.0029 ±0.0067	+0.0026 ±0.0069	-0.0012 ±0.0066
ANP + SAM	**-0.0051 ±0.0039**	**-0.0047 ±0.0043**	**-0.0070 ±0.0050**	-0.0019 ±0.0057	-0.0023 ±0.0054
ANP, random-subject split*	**-0.0198 ±0.0113**	**-0.0188 ±0.0115**	**-0.0211 ±0.0119**	**-0.0164 ±0.0119**	**-0.0120 ±0.0104**
ANP, random-subject split (width	**-0.0232 ±0.0110**	**-0.0223 ±0.0107**	**-0.0226 ±0.0115**	**-0.0176 ±0.0128**	**-0.0220 ±0.0093**

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier by labels acquired (mean ± 1 SE)

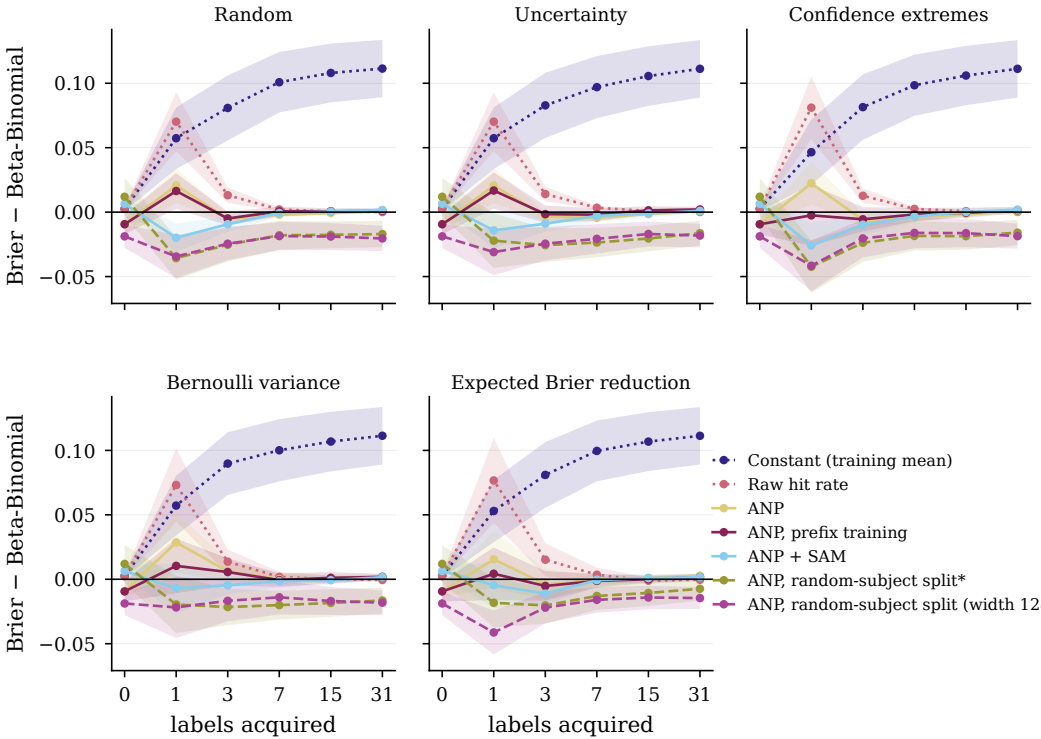
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier minus Beta-Binomial (paired, 95% CI)

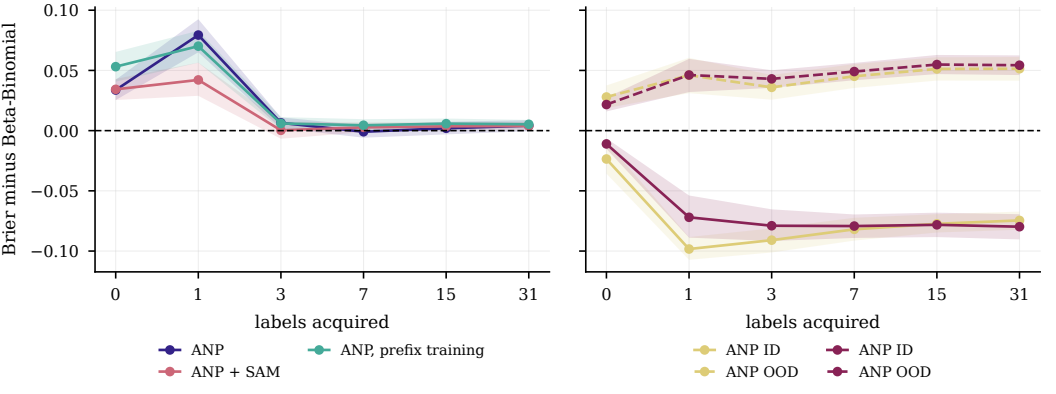
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

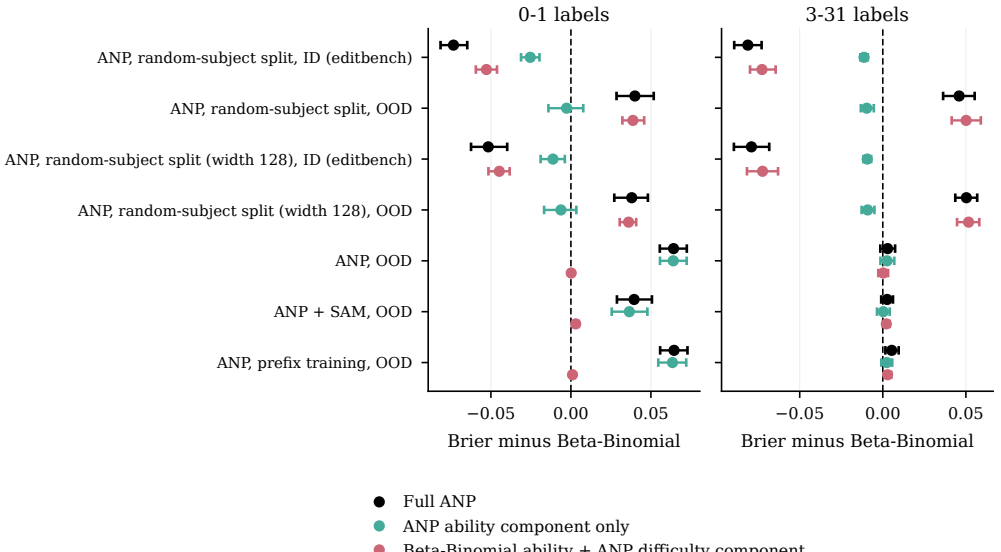
Where the ANP loses: Brier minus Beta-Binomial at each budget (test)

Left: benchmark-holdout ANPs on the three OOD test benchmarks. Right: random-subject-split ANPs on the ID benchmark editbench (solid) and the OOD benchmarks algotune + cmmlu (dashed).



Which part of the ANP fails on a new benchmark: ability or difficulty? (test)

Ability component: the mean of the ANP's predictions for the task.
Difficulty component: each item's offset from that mean.
Brier minus Beta-Binomial; negative = better. Bars: 95% task bootstrap.



Does the ANP's loss grow with subject ability?

One point per task. Ability: the subject's accuracy on the task's acquisition pool (never scored).
Shaded: 5-95% of training-task accuracy (0.00-0.87). No test or validation subject appears in training.

